Web Development Project: Spotify-Inspired Web Application

Course / Program: BSIT / IT Extension Project
Target Output: Interactive Single-Page Application (SPA) / Web Interface

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1. Project Overview & Objective

The objective of this project is to build a responsive, functional, and visually compelling web application inspired by **Spotify**. Students will demonstrate proficiency in modern frontend layout techniques, media playback controls, responsive UI design, and user experience (UX) aesthetics.

This project reinforces modern layout practices (CSS Grid/Flexbox), component structuring, event handling, media playback integration, and mobile responsiveness.

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2. Project Requirements & Specifications

A. Core Features & Layout Structure

1. **Sidebar Navigation (Fixed / Sticky):**
    - Home, Search, Library navigation links.
    - Playlist creation shortcut or saved playlist links.
    - College/Barangay extension branding or logo integration.
2. **Main Content Area:**
    - **Hero / Featured Banner:** Dynamic welcome banner (e.g., "Good morning," "Featured Tracks").
    - **Media Cards Grid:** Category cards or album grids (e.g., "Recently Played," "Top Hits," "Featured Podcasts").
    - **Card Interactions:** Hover states with play buttons and subtle scale-up effects.
3. **Persistent Bottom Audio Player Bar:**
    - **Track Info:** Album cover thumbnail, track title, and artist name. (10 songs)
    - **Playback Controls:** Play/Pause button, Previous, Next, Shuffle, Loop.
    - **Progress Bar:** Functional playback timeline slider with elapsed and remaining time.

    - **Volume Control:** Volume slider with mute/unmute capabilities.
4. **Dynamic Search / Filter (Bonus / Advanced):**
    - Real-time search filter for tracks or categories using JavaScript.

B. Technical & Responsiveness Guidelines

- **Responsive Breakpoints:**
    - **Desktop (`$\ge 1024\text{px}$`):** Full three-column view (Sidebar, Main Content, Track Info).
    - **Tablet (`$768\text{px} - 1023\text{px}$`):** Collapsible or icon-only sidebar; streamlined audio bar.
    - **Mobile (`$< 768\text{px}$`):** Bottom navbar replacing sidebar; stackable audio bar; touch-friendly controls.
- **UI/UX "Spotify Feel":**
    - **Color Palette:** Dark theme primary (#121212, #181818), Accent Green (#1DB954), Muted Text (#B3B3B3). (Based on your own preference)
    - **Typography:** Clean sans-serif font stack (e.g., Circular, Inter, system-ui).
    - **Animations:** Smooth hover transitions on cards, play buttons, and navigation links.

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3. Submission Deliverables

1. Source code hosted on GitHub / GitLab or delivered in a single HTML/CSS/JS file structure.
2. Live demo link (GitHub Pages, Vercel, or Netlify deployment).
3. Live Q&A / System Demonstration defense.

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4. Comprehensive Evaluation Rubric

Criteria	Exemplary (90–100%)	Proficient (75–89%)	Developing (60–74%)	Needs Improvement (<60%)	Weight
1. Layout & Architecture	Implements clean layout using CSS Grid and Flexbox; sidebar, main dashboard, and bottom player bar are aligned without overlapping issues.	Layout structure is complete with minor alignment or spacing inconsistencies across sections.	Basic structure exists, but layout breaks or overlaps on standard desktop viewports.	Disorganized layout; missing key structural components (sidebar, player bar, or grid).	**25%**

Criteria	Exemplary (90–100%)	Proficient (75–89%)	Developing (60–74%)	Needs Improvement (<60%)	Weight
2. Responsiveness & Adaptability	Seamlessly adapts across Desktop, Tablet, and Mobile views; uses media queries or fluid CSS; controls remain touch-friendly on mobile.	Responsive across major breakpoints with minor horizontal overflow or scaling issues on mobile.	Partially responsive; requires horizontal scrolling on mobile devices or unusable layout shifts.	Non-responsive; layout breaks completely on small screens or non-desktop viewports.	**25%**
3. "Spotify Feel" & UI/UX Design	Perfectly captures Spotify dark aesthetics, green accents, typography, smooth hover effects, card scaling, and audio bar polish.	Clean UI adhering to Spotify palette, but lacks smooth hover animations or refined micro-interactions.	Dark theme implemented, but colors, fonts, or iconography feel inconsistent with standard audio streaming UX.	Poor visual presentation; incorrect color schemes, unformatted text, or default browser aesthetics.	**20%**
4. Functionality & Playback Logic	Audio play/pause toggles work seamlessly; progress bar and volume control reflect functional states; smooth media interaction.	Audio plays and pauses correctly; minor bugs in timeline sync, volume control, or UI feedback states.	Media controls function partially (e.g., play button works, but no progress bar or track updates).	Non-functional player; buttons do not trigger events or play audio clips.	**20%**
5. Code Quality & Standards	Semantic HTML5 tags, structured modular CSS/Tailwind classes, clean JavaScript, zero console errors.	Well-structured code with minor redundancy or occasional unused CSS/JS blocks.	Unorganized code structure; heavy reliance on inline styles; non-semantic markup.	Disorganized code; multiple broken scripts or unformatted files.	**10%**

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5. Defense & Q&A Evaluation Checklist

When evaluating the **Q&A / Oral Defense** section of this project, consider:

- **Technical Justification:** Can the student explain how CSS Grid/Flexbox and JS audio events were implemented?
- **Problem-Solving Ability:** How did the student handle responsive design challenges on mobile screens?
- **Product Vision:** How effectively did the student maintain user experience consistency with real-world streaming applications?